

Gerald Darryl Joseph Manurung

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Informatics student (GPA 3.90/4.00) specializing in Security Engineering and Full-Stack Development, with hands-on experience building end-to-end encrypted (E2EE) systems, AI-driven SOC/SIEM platforms, and NIST CSF compliance tools. Proficient in vulnerability analysis, applied cryptography (AES/RSA), and security operations (Wazuh, Wireshark, Burp Suite), with competitive results including 2nd place in Digital Forensics (PU CTF) and 3rd place at Garuda Hacks 7.0. Google Student Ambassador (top 2.5% of 81,000+ applicants) and Supervisor of COMPSPHERE 2026, leading a 100+ member cross-functional team across four large-scale IT sub-events.

Projects

Team Lead & Full Stack Developer

Project GUARDIAN | AI-Augmented SOC/SIEM Platform for Fintech API Security

- Architected an async, out-of-band telemetry pipeline (Redpanda, Vector, OpenSearch) to ingest and normalize API traffic at a simulated 5M+ transactions/day, eliminating alert-fatigue-inducing raw log volume.
- Engineered a three-model ML detection triad: XGBoost log classification, Isolation Forest rate/payload anomaly baselining, and CUSUM/LSTM-autoencoder drift detection. Each model is tasked to surface behavioral anomalies and low-and-slow data exfiltration that signature-based rules miss.
- Built Guardian Pulse, a Next.js/TypeScript SOC dashboard delivering real-time threat scoring, entity-level case-file drilldowns, an alert feed, and one-click Freeze-to-PDF compliance snapshots.

Lead Developer

CrowdFlow | Event Ticketing & Venue Management Platform

- Architected a Next.js/TypeScript platform for event discovery, seat-level ticket purchasing, and role-based access control across buyer, organizer, auditor, and admin roles.
- Engineered a Go backend with PostgreSQL, building secure JWT authentication, Midtrans payment integration, and an interactive seat-map/venue-layout editor with per-tier pricing.
- Orchestrated a document-verification and compliance workflow (KTP/NPWP/NIB auditing, payout approvals) alongside S3-backed public/private object storage for organizer and event assets.

Full Stack Developer

WHISPXR | An End-to-End Web-based Messaging Application

- Engineered a secure, E2EE messaging platform using Next.js and Firebase, serving 50+ users.
- Implemented hybrid AES-256 and RSA client-side encryption to guarantee message privacy from server and third-party access.
- Architected a real-time system using Firebase to maintain low-latency message delivery for concurrent connections
- Optimized backend performance via lightweight caching and idempotent flows, reducing server latency and resolving race conditions.

Full Stack Developer

Assetra | AI-Powered Cybersecurity Governance & Auditing Platform

- Architected a Next.js/Typescript platform for real-time risk scoring, URL integrity monitoring, and role-based access control (RBAC).
- Engineered Express.js backend APIs and structured Supabase database schemas with secure authentication.
- Orchestrated LLM data pipelines to transform complex NIST CSF 2.0 compliance data into actionable UI visualizations.

Experiences

Google Student Ambassador Indonesia

Student Ambassador

April 2026 - Present • Hybrid

- Appointed as a Google Student Ambassador, placing in the top 2.5% of an 81.000+ national applicant pool, to evangelize Google's mission and technology on campus.
- Promoted Google Cloud and Gemini AI by leading hands-on workshops for 100+ students, fostering digital literacy.
- Collaborated with a national network of student leaders to implement Google technologies in academic research and projects.

Computing Atmosphere (COMPSPHERE) 2026

Supervisor

February 2026 - Present • Cikarang, West Java

- Oversee and lead the strategic execution of COMPSPHERE, the largest annual IT event hosted by President University Faculty Association (PUFA) of Computer Science, managing four distinct sub-events: Hacksphere, Festsphere, Exposphere, and Talksphere.
- Directing a cross-functional workforce of over 100 committee members across technical and logistics departments, ensuring synchronized execution for an event series expected to draw more than 500 attendees.
- Secure external partnerships and guest speakers to enhance the event's professional prestige and networking opportunities.

Computing Atmosphere (COMPSPHERE) 2025

Lead Hackathon Organizer

February 2025 - November 2025 • Cikarang, West Java

- Led the COMPSPHERE first-ever hackathon, overseeing end-to-end planning and execution of a 48-hour national-level competition,
- Helped establish the first hackathon format in COMPSPHERE history, setting a foundation for future large-scale tech competitions
- Managed a core team of 6 hackathon organizers and coordinated event operations for 87 competing participants from multiple universities and institutions
- Strengthened capabilities in leadership, large-scale event management, cross-team coordination, and high-pressure problem solving

Technical Competitions

PU CTF: Digital Forensics | *Rank: 2nd out of 30 Teams*

April 2026 • Cikarang, West Java

- Performed incident reconstruction and forensic artifact recovery by analyzing disk images and network traffic under time-sensitive constraints.

PU CTF: Ethical Hacking | *Rank: 13th out of 38 Teams*

October 2025 • Cikarang, West Java

- Executed vulnerability assessment and exploitation strategies, demonstrating proficiency in web security, privilege escalation, and testing methodologies.

Garuda Hacks 7.0 | 3rd Place in the Agriculture & Food Systems track

July 2026 • Universitas Multimedia Nasional, Tangerang

- Architected an agricultural market-signaling Telegram bot, achieving 3rd place, demonstrating proficiency in Python, Supabase, and architecture.

Education

President University

Sarjana Komputer (S.Kom.)

August 2024 - Present • Cikarang, West Java • Expected Graduation: December, 2027

- Cumulative GPA: 3.90 / 4.00
- Relevant Coursework: Data Structure and Algorithms, Ethical Hacking, Network Security, Artificial Intelligence, Software Engineering, Cryptography and Data Security, Operating System Design, and Security Risk Management and Audit.
- Research and Analysis: Conducting academic research on the correlation between social habits and academic performance, utilizing data collection and statistical analysis techniques.

Technical Skills

Languages: C++, Python, Go, Typescript, Bash, Java, PHP, SQL, Javascript

Web Development: React, Next.js (App Router), Express.js, Laravel, Tailwind CSS

Cybersecurity: Wireshark, Wazuh, Burp Suite, NIST CSF, Cryptography, Web Security Best Practices

Tools & DevOps: Git, Docker Compose, Nginx, Google Cloud Platform, n8n

Databases: MySQL, PostgreSQL, Redis, Supabase

Certifications

Introduction to Cybersecurity (Cisco) • Certified Junior Web Developer (BNSP)

Languages

English (Fluent) • Indonesian (Native)